Influence – Changing the Model

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Identifying “Influential” Data Points

- Thus far we have discussed how we detect unusual values in x (levers) and unusual values in y (outliers).

- Combining these two factors, we can measure the “influence” of individual data points.
  - Influence refers to the affect that a data point has on our model when it is present compared to when it is absent.

  - An ***influential*** data point can change our parameter estimates in either their magnitude, their statistical significance, or both.

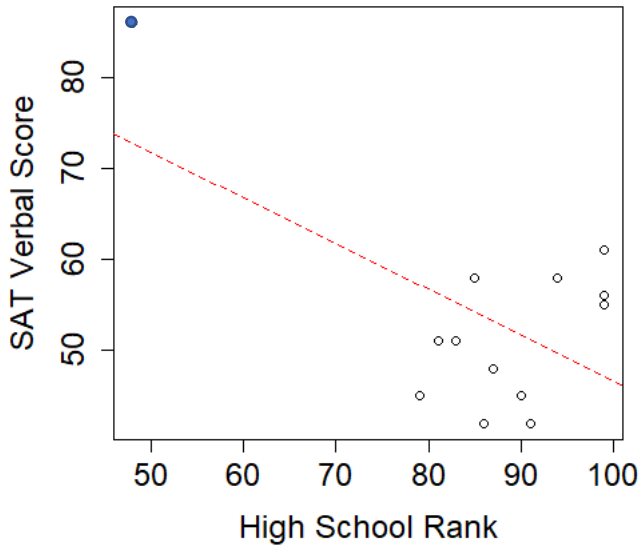
  - As such, influential data points are often excluded (and models re-run) because we want our model to reflect ***all*** the data, not be swayed by a single strange point.

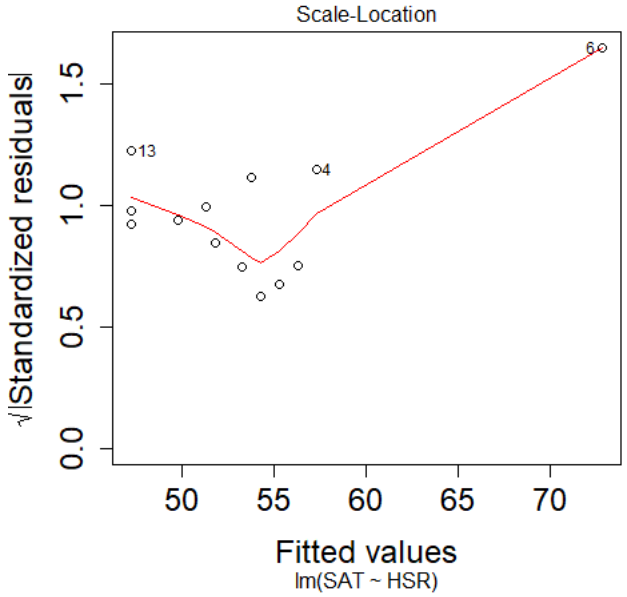
Calculating Influence

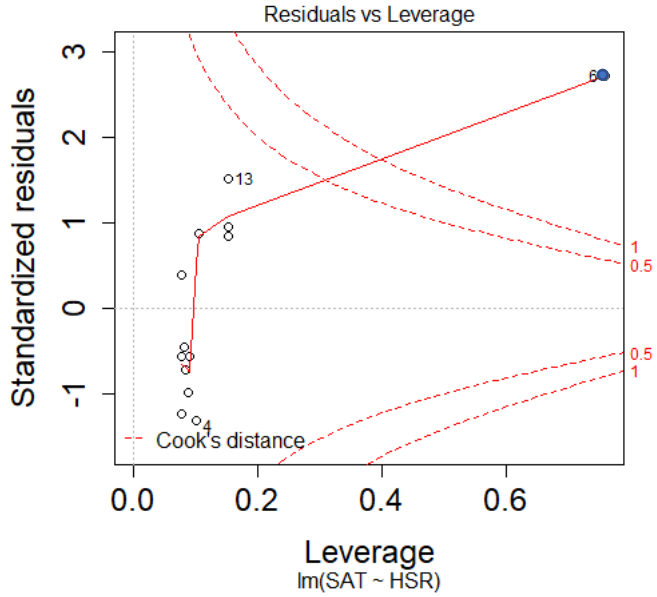
- **Cook's Distance** is the most common measure of influence in regression.
  - $D_i = \frac{e_i^2}{PA(MSE)}\left[\frac{h_i}{(1-h_i)^2}\right]$

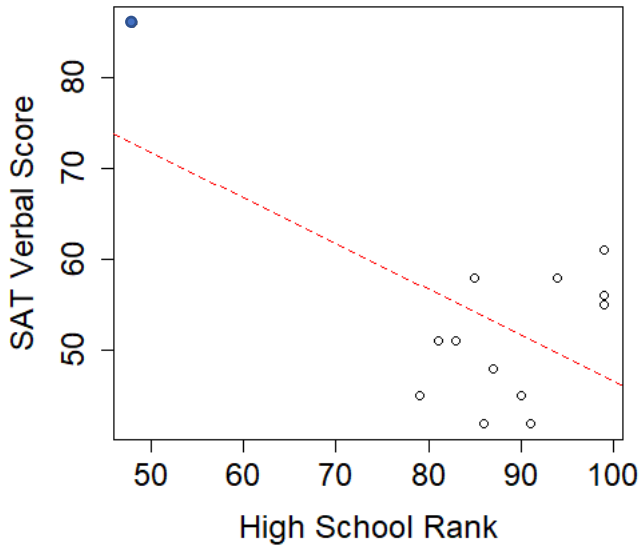
- Note that Cook's Distance, D_i, tells us about the influence a data point has on a model it were to be deleted, but all of the component parts (e_i, h_i, PA, MSE) come from the model with that data point included.
  - This means that we don't need to tediously re-run model after model.

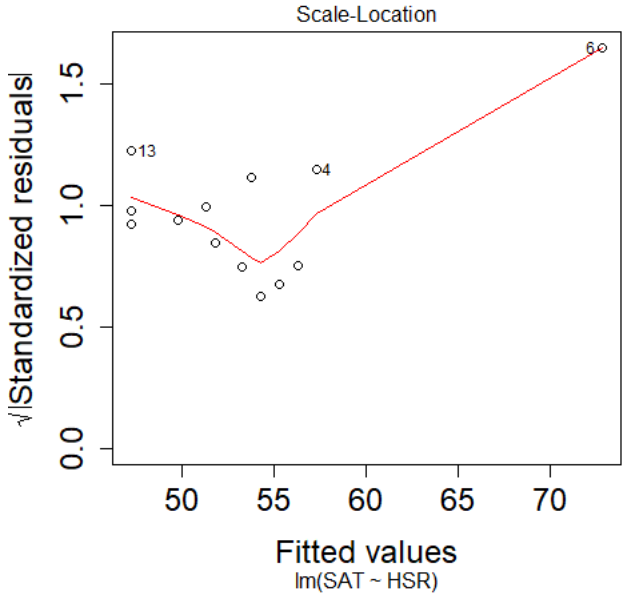
- Cook's distance combines outlier distance (on the y-axis) with leverage (on the x-axis) to get a combined measure of **<u>influence</u>**.

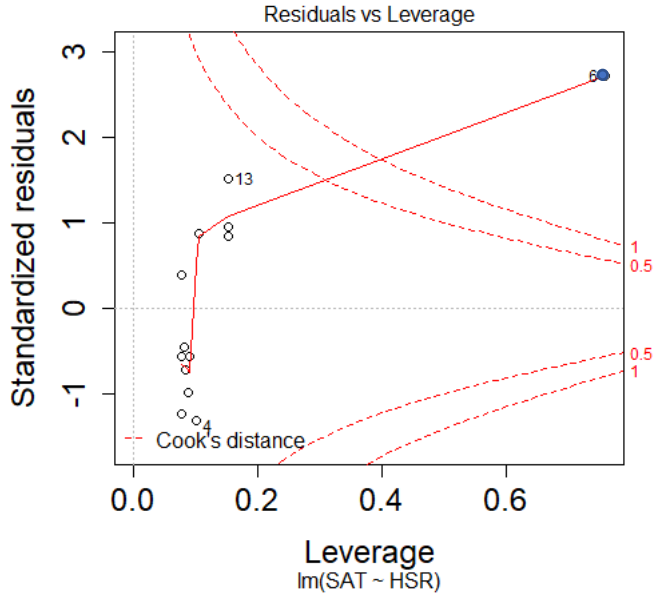

SAT Verbal Score
50
60
70
80
50
60
70
80
90
100
High School Rank


Scale-Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
13
4
6
50
55
60
65
70
Fitted values
lm(SAT ~ HSR)


Residuals vs Leverage
Standardized residuals
-1
0
1
2
3
13
6
4
1
0.5
0.5
1
Cook's distance
0.0
0.2
0.4
0.6
Leverage
lm(SAT ~ HSR)







Influential data point included:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  96.8852    18.6356   5.199 0.000295 ***
HSR          -0.5009     0.2138  -2.343 0.038946 *
```

Influential data point **removed**:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   8.2919    22.6394   0.366   0.7218
HSR           0.4776     0.2525   1.892   0.0878 .
```